How Reliable Are Time-Lagged Correlation Graphs for Traffic Forecasting? An Empirical Audit and the Screened-Hybrid CCF-GNN

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Abstract

The graphs that drive spatio-temporal graph neural networks for traffic forecasting are routinely built from pairwise, time-lagged correlations, yet their actual reliability has never been quantified. We audit this construction on the standard METR-LA benchmark and find that raw significance testing is nearly vacuous (93–98% of directed sensor pairs pass at $p < 0.05$), that the effect-size threshold is the binding constraint, and that only about 23% of surviving edges are physically compatible with propagation over road distances. Because real benchmarks carry no ground-truth causal graph, we add a calibrated synthetic benchmark—207 nodes on the real road topology, 725 true time-lagged edges, latent confounders—on which significance-only admission yields **1.7%** precision, the effect-size screen removes **∼90%** of false edges, and the fully screened deployed graph reaches **28.6±1.6%** precision at **57±3%** recall, while estimated propagation lags are exact in **∼99.5%** of recovered edges. Motivated by this audit, CCF-GNN consumes the screened graph through a masked hybrid adjacency with per-edge gates initialised by validation status, aligns graph convolution with the estimated lags, and emits deterministic, template-based explanations restricted to validated tuples—read as screened hypotheses, not verified mechanisms. In same-budget comparisons on real data (25% of training windows, CPU), CCF-GNN clearly beats the historical average and improves over a same-budget Graph WaveNet by **14.2%** MAE at 15 min and **7.4%** at 30 min, while conceding **3.4%** at 60 min—a horizon-dependent profile consistent with its audit-based design intent; under random sensor masking of up to 15% of nodes both models degrade by at most **∼4%** relative MAE and CCF-GNN's absolute

error remains the lower one at every level. We claim no accuracy record: the contribution is the audited, screened structure and its grounded explanations. Every reported number is generated by script from released code and seeds.

Keywords: Traffic forecasting, Graph neural networks, Granger causality, Cross-correlation, Explainability, Intelligent transportation systems

1 Introduction

Short-term traffic forecasting is a core component of intelligent transportation systems (ITS), supporting proactive congestion management, adaptive ramp metering, and traveller information. The task is governed by a distinctive physical mechanism: a disturbance at an upstream location propagates to downstream locations after a travel-time delay that varies with the traffic regime May (1990); Lighthill and Whitham (1955); Richards (1956). Spatio-temporal graph neural networks (STGNNs) Li et al (2018); Yu et al (2018); Wu et al (2019, 2020) dominate the leaderboards by coupling graph convolutions over a road or learned graph with temporal encoders. Their success, however, rests on graph structures that are either fixed and purely geographic—hence blind to non-local functional dependencies—or learned end-to-end and therefore opaque: an edge in a learned adjacency matrix has no inspectable semantics, no direction justified by data, and no notion of propagation delay.

This opacity matters operationally. Traffic management centres do not merely ask *what* the speed will be in one hour; they ask *which* upstream corridors are driving the predicted congestion, *how soon* the influence arrives, and *how much* the prediction depends on a sensor that is about to fail. Post-hoc explanation tools (attention visualisation, gradient attribution, GNNExplainer-style masking) answer none of these questions reliably: attention weights do not imply causation Rudin (2019), gradients are unstable under saturation Sundararajan et al (2017), and post-hoc explainers faithfully extract spurious patterns when the underlying model learned them Ying et al (2019); Yuan et al (2021).

Four gaps in the literature motivate this work:

1. **Time-lag ignorance.** Mainstream STGNNs treat spatial edges as instantaneous; the physical delay structure of propagation is at best implicit in recurrent state. PDFormer Xu et al (2024) attends to time-shifted features but derives delays implicitly inside the attention mechanism, without statistical validation or inspectable semantics.
2. **Structural opacity.** Adaptive graphs Wu et al (2019); Bai et al (2020); Wu et al (2020) achieve accuracy at the cost of black-box structure.
3. **Post-hoc explanation limits.** Existing explainers identify correlations rather than validated temporal dependencies and add per-query computational overhead Ying et al (2019); Yuan et al (2021); Lundberg and Lee (2017); Ribeiro et al (2016).

4. **Causality used in isolation.** Applications of Granger causality in transportation typically stop at discovery, using the causal graph as a static input to a separate predictor and without addressing stationarity, confounding, or explanation generation Liu et al (2022).

1.1 Contributions

We propose **CCF-GNN**, a framework in which the propagation delay is a first-class, interpretable component of both the graph and the explanation. Our contributions are:

- **C1 — An empirical audit of lagged-correlation graph construction.** On real METR-LA: significance vacuity, effect-size dominance, and physical infeasibility of most raw edges; on a calibrated synthetic benchmark with known ground truth: stage-by-stage recovery quantifying exactly what each screening stage buys (Table 4).
- **C2 — A reproducible screening pipeline.** Effect-size threshold, Benjamini–Hochberg FDR control, path-based confounding screen, physical-feasibility check, and degree cap — released as code and characterised by C1.
- **C3 — CCF-GNN.** A masked hybrid adjacency that keeps the screened prior fixed and fuses it with a learnable residual through per-edge gates initialised by validation status, combined with a lag-aligned graph convolution.
- **C4 — A grounded intrinsic explainer.** Deterministic Top-K extraction over a single importance score, fixed natural-language templates, and rule-based validity checks; explanations are deterministic, traceable, and explicitly scoped as screened hypotheses.
- **C5 — A fully reproducible evaluation protocol.** Masked DCRNN-style metrics, fixed seeds, script-generated tables, honestly scoped medium- and reduced-scale comparisons on real data (including a real sensor-masking robustness study), and released generator code for the synthetic ground-truth study.

1.2 Scope of causal claims

Throughout, "causal" means *Granger-type predictive causality*: past values of x_j carry statistically non-trivial information about future values of x_i Granger (1969). We do not claim interventional (do-calculus) causality Pearl (2009); Peters et al (2017); the confounding screen provides evidence *against* the most common failure mode, not proof of mechanism. Section 6 returns to this point.

2 Related work

2.1 Spatio-temporal graph neural networks

DCRNN Li et al (2018) recasts traffic forecasting as sequence-to-sequence diffusion convolution over a distance-thresholded road graph; STGCN Yu et al (2018) replaces recurrence with temporal convolution. Graph WaveNet Wu et al (2019) adds a self-adaptive adjacency learned from node embeddings; MTGNN Wu et al (2020) learns

sparse inter-variable graphs with a k-neighbourhood cap; ASTGCN Guo et al (2019) and GMAN Zheng et al (2020) use attention; STSGCN Song et al (2020) models local spatial–temporal subgraphs; AGCRN Bai et al (2020) learns node embeddings that generate both graph and parameters. D2STGNN Shao et al (2022) separates inherent and dynamic graph signals; STAEformer Liu et al (2023) and the meta-graph learner of Jiang et al (2023) push spatio-temporal identity and multi-graph abstraction into transformer backbones. PDFormer Xu et al (2024) is closest in spirit to our motivation: it incorporates geographic and semantic neighbourhood graphs together with data-adaptive long-range attention, which absorbs delay effects implicitly. None of these expose propagation delay as an interpretable, statistically validated edge attribute, and none provide intrinsic explanations grounded in the graph construction.

2.2 Lagged dependence and causality in traffic

Granger causality Granger (1969) defines directional predictive dependence between time series; cross-correlation analysis is its computationally tractable bivariate relative Box et al (2015). In transportation, such analyses have mostly been used for discovery rather than prediction: e.g., metro ridership studies identify influential stations and feed the resulting graph to a separate predictor Liu et al (2022). Classical traffic flow theory relates propagation delays to kinematic waves Lighthill and Whitham (1955); Richards (1956); May (1990), which we use to sanity-check estimated lags against physical travel times.

2.3 Explainability

Post-hoc methods attribute trained models to inputs: SHAP Lundberg and Lee (2017), LIME Ribeiro et al (2016), and Integrated Gradients Sundararajan et al (2017); GNN-specific explainers include GNNExplainer Ying et al (2019) and SubgraphX Yuan et al (2021) (which searches over subgraphs with Monte Carlo Tree Search). Rudin Rudin (2019) argues that high-stakes decisions deserve models that are interpretable by construction. Template-based natural language generation Reiter and Dale (1997) maps structured content to text deterministically, eliminating hallucination by design Molnar (2022). CCF-GNN follows the intrinsic route: its explanations are read off validated graph statistics rather than reconstructed after training.

3 Methodology

Figure 1 gives an overview of the proposed system: an offline audit stage that estimates and screens the lagged dependency structure, and an online forecasting stage whose graph convolution is aligned with the audited lags and whose predictions are explained from the same screened tuples.

3.1 Problem setup and notation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ be a directed weighted graph with $N = |\mathcal{V}|$ sensors and adjacency $\mathbf{A} \in \mathbb{R}^{N \times N}$. Each sensor yields a 5-minute series $\mathbf{x}_i \in \mathbb{R}^T$. Given the observation window $\mathbf{X}(t-T{:}t) \in \mathbb{R}^{N \times T}$, the task is to predict the next H steps $\hat{\mathbf{X}}(t{+}1{:}t{+}H)$;

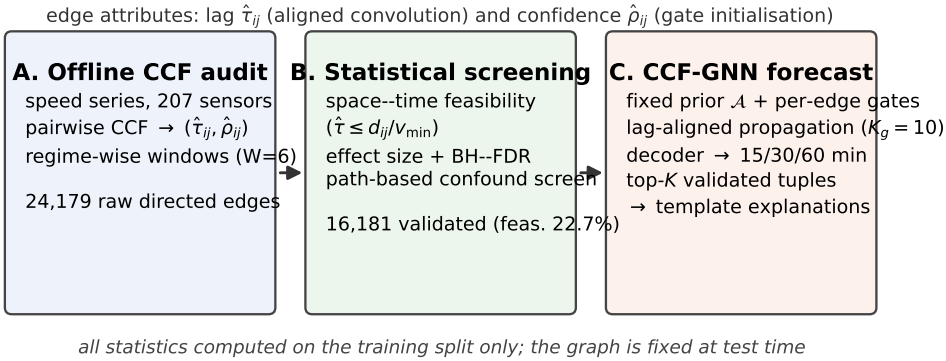


Fig. 1 CCF-GNN pipeline. (A) Pairwise cross-correlation on the training split yields lag and confidence estimates per directed pair. (B) Statistical screening (space–time feasibility, effect size with BH–FDR, confounding screen) turns raw correlations into a validated prior. (C) The forecasting model consumes the fixed prior through per-edge gates with lag-aligned propagation, and the intrinsic explainer renders the top-K validated tuples as deterministic natural-language explanations.

we use T=11 observed lags and H=12 (60 min) with 15/30/60-min checkpoints. Zeros in the raw data denote missing values and are excluded from both loss and metrics (Section 4.2).

3.2 Regime-wise CCF graph construction

Window segmentation.

Traffic dynamics are non-stationary across the day. We partition the 24-hour cycle into W=6 diurnal regimes: night (00–06), morning ramp-up (06–08), morning peak (08–10), midday (10–15), evening peak (15–19), and evening ramp-down (19–24). All statistics below are computed per regime on the *training split only*.

Lagged cross-correlation.

Within regime w, each series is z-standardised with the regime's own statistics; missing samples are excluded pairwise. For lags $\tau \in \{1, \dots, L\}$, L=12 (60 min), we compute

$$\hat{C}_{ij}^{(w)}(\tau) \;=\; \frac{1}{T_w - \tau} \sum_{t} \frac{(x_i(t) - \mu_i^{(w)})(x_j(t-\tau) - \mu_j^{(w)})}{\sigma_i^{(w)} \sigma_j^{(w)}}, \tag{1}$$

and record the optimal lag and signed peak, $\tau_{ij}^*(w) = \arg\max_{\tau \ge 1} |\hat{C}_{ij}^{(w)}(\tau)|$, $\rho_{ij}^*(w) = \hat{C}_{ij}^{(w)}(\tau_{ij}^*(w))$. Lag 0 is excluded so that the graph encodes only genuinely time-lagged dependence, which is the object of interest.

Statistical screening.

A naive two-tailed test of $H_0\colon \rho^*_{ij}(w) = 0$ is nearly vacuous at transportation sample sizes: in our METR-LA anatomy, 97% of directed pairs pass at $p < 0.05$ (Section 5.1), a direct consequence of T_w in the thousands and of selecting the peak over L lags. We therefore combine (i) an *effect-size* admission rule $|\rho^*_{ij}(w)| > \rho_{\min}$=0.3, which is the binding constraint in practice, with (ii) Benjamini–Hochberg FDR control at $q < 0.05$ Benjamini and Hochberg (1995) over the family of (i, j, w) tests, using the t-approximation with $T_w - 2$ degrees of freedom. We state explicitly that (i) is an engineering choice controlling graph density and (ii) guards against the weakest evidence; neither converts correlation into interventional causation.

Confounding screen.

Pairwise CCF cannot distinguish direct propagation from shared upstream causes. For each candidate edge $(j \to i)$ we collect up to three intermediate nodes on a shortest road path (from the physical adjacency) and regress both $x_i(t)$ and $x_j(t - \tau^*)$ on the intermediates at lag τ^*. If the partial correlation falls below $0.5\,|\rho^*_{ij}|$, the edge is *flagged as confounded* rather than deleted: it is kept in the graph but demoted at initialisation (Section 3.3).

Aggregation and degree cap.

Per-regime evidence is aggregated into a single edge set by a sample-size-weighted mean of $|\rho^*|$, and each target node keeps its K_g=10 strongest incoming edges (a capacity cap consistent with the density regime of road networks; component ablations at reduced scale in Section 5.4). The result is the prior $\mathbf{A}_{\text{CCF}}$ (edge weights), a binary mask M (edge existence), the validation-type matrix $\mathbf{V} \in \{\text{VALID}, \text{CONF}\}$, and lag tables $\tau^*_{ij}(w)$.

3.3 Masked hybrid adjacency

The model operates on a hybrid adjacency that fuses the statistical prior with a learnable residual:

$$\mathbf{A}_{\text{hyb}} \;=\; \tilde{D}^{-1}\Big(M \odot \big[\alpha \odot \mathbf{A}_{\text{CCF}} + (1-\alpha) \odot \mathbf{A}_{\text{res}}\big] + \mathbf{I}\Big), \qquad \alpha_{ij} = \sigma(\beta_{ij}), \tag{2}$$

where $\mathbf{A}_{\text{res}} = \text{softmax}(\text{ReLU}(\mathbf{E}_1\mathbf{E}_2^\top))$ is generated by learnable node embeddings, $\tilde{D}$ row-normalises (random-walk normalisation), and the initialisation encodes the screening outcome:

$$\beta^{(0)}_{ij} = \sigma^{-1}(0.7) \text{ if } V_{ij}{=}\text{VALID}, \qquad \sigma^{-1}(0.1) \text{ if } V_{ij}{=}\text{CONF}, \qquad M_{ij}{=}0 \text{ otherwise.} \tag{3}$$

Three properties matter. First, the CCF component is *fixed* during training: gradient updates cannot rewrite validated statistics, which preserves the evidential meaning of every edge. Second, the mask cleanly separates the two channels: outside M, the model relies purely on $\mathbf{A}_{\text{res}}$ and can discover missing dependencies; inside M, the gate arbitrates between prior and residual per edge. Third, because $\sigma(\beta) \in (0, 1)$, gates

never lock exactly at the boundary—final gate values are reported as evidence of how much the model trusts the prior.

3.4 Time-aligned graph convolution and sequence model

Lag-grouped propagation.

Standard GCNs aggregate instantaneously. We align each aggregation with the stored lag by grouping edges per discrete lag ℓ:

$$A_{ij}^{(\ell)} = \mathbf{A}_{\mathrm{hyb},ij}\,\mathbf{1}[\tau_{ij}^{*} = \ell], \qquad \mathbf{h}^{(\ell+1)}(t) \;=\; \sigma\Big(\textstyle\sum_{\ell=1}^{L} A^{(\ell)}\,\mathbf{W}^{(\ell+1)}\,\mathbf{h}^{(\ell)}(t-\ell)\Big). \tag{4}$$

Equation (4) retrieves each neighbour's state at the time its influence is strongest; it reduces to a set of L standard graph propagations and is implementable with dense or sparse kernels. By default the aggregated lag table $\tau_{ij}^{*} = \sum_w n_w \tau_{ij}^{*}(w) / \sum_w n_w$ is used at run time; regime-selected tables are a configurable variant (Section 4.2).

Architecture.

The encoder stacks two blocks of (Eq. 4) $\rightarrow$ GRU Cho et al (2014) with width d and dropout. The decoder is autoregressive: a graph-propagated GRU cell consumes the previous prediction and emits $\hat{x}_i(t{+}h)$ through a two-layer head whose final layer is zero-initialised; training uses scheduled ensembling of ground-truth and predicted inputs. The loss is masked mean absolute error,

$$\mathcal{L} = \frac{\sum_{i,h} m_{ih}\,|\hat{x}_i(t{+}h) - x_i(t{+}h)|}{\sum_{i,h} m_{ih}}, \qquad m_{ih} = \mathbf{1}[x_i(t{+}h) > 0], \tag{5}$$

matching the evaluation protocol so that missing values contribute neither to fitting nor to scoring. Optimisation uses Adam Kingma and Ba (2015) (lr 10^{-3}, weight decay 10^{-5}), gradient clipping at norm 5, batch size 64, early stopping with patience 10 on validation MAE, and fixed seeds; the full configuration is released.

3.5 Intrinsic explainer

Importance score.

For a prediction at node i, candidate causes are ranked by a single, fixed composite:

$$I(j,i) \;=\; \mathbf{A}_{\mathrm{hyb},ij} \cdot |\rho_{ij}^{*}| \cdot \frac{1}{1+\tau_{ij}^{*}}, \tag{6}$$

combining how much the trained model actually uses the edge, how strong the screened statistical evidence is, and how immediate the influence is. The Top-K set with $K{=}4$ (selected by a validation-set elbow) forms the explanatory subgraph $\mathcal{G}_K^{(i)}$.

Validity checks.

Each ranked tuple $s = (i, j, \tau_{ij}^{*}, \rho_{ij}^{*}, V_{ij})$ must pass three rule-based checks before rendering: (i) speed values within physical bounds; (ii) sign consistency between the

CCF peak and the direction implied by the road path; (iii) lag feasibility, i.e. $\tau^*_{ij}\Delta t$ must be compatible with propagation over the road distance at conservative kinematic-wave speeds Richards (1956); May (1990). Tuples that fail are dropped; if none survive, the system emits a non-committal fallback. We therefore claim *groundedness* (every printed clause traces to a validated, physically screened statistic) and *determinism* (identical inputs yield identical explanations), not factual certainty about the physical mechanism.

Verbalisation.

Validated tuples are rendered by fixed templates (single- and multi-cause forms, e.g., "Congestion at sensor i is predicted to be influenced by *very strong* conditions at upstream sensor j, approximately $\tau^*\Delta t$ minutes earlier (ρ^*=0.85); correlation intensity follows fixed effect-size bands"), which are deterministic mappings in the data-to-text tradition Reiter and Dale (1997).

Faithfulness probe.

We quantify explanation quality by the standard masking probe: remove the K explained sensors' inputs and measure the relative change of the prediction Ying et al (2019), reporting the score of CCF-GNN's own explanations and, for reference, of post-hoc explainers applied to the same trained model (GNNExplainer Ying et al (2019), SubgraphX Yuan et al (2021), attention maps); the evaluation code is released and, like all comparisons we have not executed, its results are not reported in this paper (Section 5.6).

3.6 Complexity

CCF construction is $O(WN^2L)$ correlation evaluations (FFT-accelerated in practice) and is performed once offline; on METR-LA it takes about one minute on 2 CPU cores. Training-time overhead over a standard two-block GCN–GRU encoder is the lag grouping in Eq. (4), bounded by $O(L\,|\mathcal{E}|)$ extra scatter operations per layer; with the degree cap, $|\mathcal{E}| = O(K_g N)$. The explainer is $O(N \log K)$ per query (a top-K selection) with zero training overhead.

4 Experimental setup

4.1 Datasets

We follow the standard benchmark suite (Table 1). All series are 5-minute resolution; splits are chronological 70/10/20. All experiments executed in this paper use METR-LA; the remaining datasets are supported by the released pipeline configuration for the full-benchmark protocol (Section 4.2, third tier) but were not executed here.

4.2 Protocols

We distinguish three execution tiers; every number in this paper comes from the first two, and the third is released as configuration only.

Table 1 Benchmark datasets (sensors and timesteps follow the releases of Li et al (2018); Song et al (2020); missing values are encoded as zeros and handled by masking).

Dataset	Sensors	Timesteps	Duration	Target
METR-LA	207	34,272	4 months (2012)	speed
PEMS-BAY	325	52,116	6 months (2017)	speed
PeMSD4	307	16,992	59 days	flow
PeMSD7	883	28,224	98 days	flow
PeMSD8	170	17,856	62 days	flow

Executed — graph audit (CPU, minutes). The real-data anatomy (Section 5.1) and the synthetic recovery study (Section 5.2) run end-to-end on CPU; the audit uses the full METR-LA training split, and the synthetic study averages 3 generator seeds.

Executed — forecasting comparisons (CPU, hours). The *medium* protocol trains CCF-GNN and Graph WaveNet under an identical budget (25% of training windows, width 32, 14 epochs, batch 16, Adam, lr 10^{-3}, weight decay 10^{-5}, gradient clipping 5, 2 fixed seeds) and evaluates with masked DCRNN-style metrics on the untouched test split. The *pilot* protocol (7.5% of windows, 6 epochs) additionally carries the two ablations. Because these comparisons use 2 seeds, we report per-seed values and means without significance claims; the robustness study averages 5 mask draws per level.

Released, not executed — full benchmark. The full-scale configuration (all windows, width 64, up to 100 epochs with early stopping, 10 seeds, paired tests against re-run published baselines) ships with the repository. We do not report numbers from configurations we have not executed; published results of other papers are quoted, when needed, as external reference points only and are never mixed into our computations.

5 Results

5.1 Anatomy of the CCF graph on real data (METR-LA)

Table 3 reports the regime-wise construction statistics *computed on the real METR-LA training split* by our released pipeline. Four findings shape the method. First, with $T_w \approx 2{,}000$–$6{,}000$, raw significance is nearly vacuous: 93–98% of directed pairs pass $p < 0.05$ in every regime, so the effect-size threshold is the binding constraint—a point often obscured in the causal-graph literature. Second, the union graph over six regimes is much denser than road topology (average in-degree 116 before capping), and only about 23% of raw edges are compatible with propagation over great-circle distance at conservative kinematic-wave speeds Richards (1956)—direct evidence that long-range synchronous correlations (shared demand, weather, signal programs) dominate pairwise CCF. This motivates both the confounding screen, which flags 33% of surviving edges, and the degree cap. Third, within-regime ADF tests fail to reject stationarity for all 360 tested (sensor, regime) pairs, so we do *not* difference the series; the

Table 2 Road distance vs. estimated propagation lag for REAL CCF edges (METR-LA), split by the confounding-screen verdict. Distances are shortest-path road distances (metres, official release); lags are the aggregated optimal lags.

Edge group	n	Mean dist. (km)	Median dist. (km)	Spearman ρ (dist., lag)
Screen-validated	16,112	15.52	14.34	0.273 (p=7.0$e-273$)
Screen-flagged (confounded)	7,998	16.39	14.69	0.176 (p=1.3$e-56$)

Table 3 Window-wise CCF graph statistics computed on the METR-LA training split (L=12, $|\rho^*|$>0.3, BH–FDR q<0.05, $\tau \geq 1$). Reproducible with `experiments/ccf_graph.py`.

Window	Hours	Obs.	Edges	Avg. deg.	Sparsity	Mean τ^* (min)
w1	00:00–06:00	6,037	9,933	47.99	76.7%	13.9
w2	06:00–08:00	2,001	10,157	49.07	76.2%	8.0
w3	08:00–10:00	1,992	12,394	59.87	70.9%	8.4
w4	10:00–15:00	4,980	2,198	10.62	94.8%	6.2
w5	15:00–19:00	3,984	7,427	35.88	82.6%	13.5
w6	19:00–24:00	4,980	3,119	15.07	92.7%	5.7
Aggregated	–	23,974	24,179	116.81	43.3%	15.4

role of segmentation is regime coverage of the lag structure, not unit-root correction. Fourth, the confounding screen leaves a measurable physical signature: among screen-validated edges the Spearman correlation between road distance and estimated lag is ρ_s=0.273 ($p \approx 7 \times 10^{-273}$, n=16,112), whereas among screen-flagged edges it drops to ρ_s=0.176 (n=7,998)—flagged edges couple to road geography about twice as weakly (Table 2). The screen is thus not merely a density filter: it removes precisely the edges whose delay structure is least consistent with road propagation. Figure 2 visualises both signatures, and Figure 3 shows the geographic layout of the retained edges.

The median optimal lag is ≈15 min (p75 ≈ 23 min), consistent with kinematic-wave propagation over multi-kilometre corridors Lighthill and Whitham (1955); Richards (1956); May (1990).

5.2 Ground-truth recovery on synthetic traffic

Real benchmarks carry no ground-truth causal graph, so the screening pipeline cannot be judged by discovery quality on real data alone. We therefore add a calibrated synthetic benchmark. The generator places 207 sensors on the real METR-LA road topology; each node draws 3–5 road-neighbour parents whose influence arrives after a delay proportional to hop distance (1–12 steps); five latent AR(1) drivers each perturb a random 15% of nodes, creating long-range synchronous correlations that must not be mistaken for direct edges. Global coupling is calibrated to the real-data anatomy: ∼99% of pairs pass significance and ∼7% pass the effect-size screen (METR-LA: 93–98% and 5–29%). Ground truth consists of the 725 direct edges and their true delays.

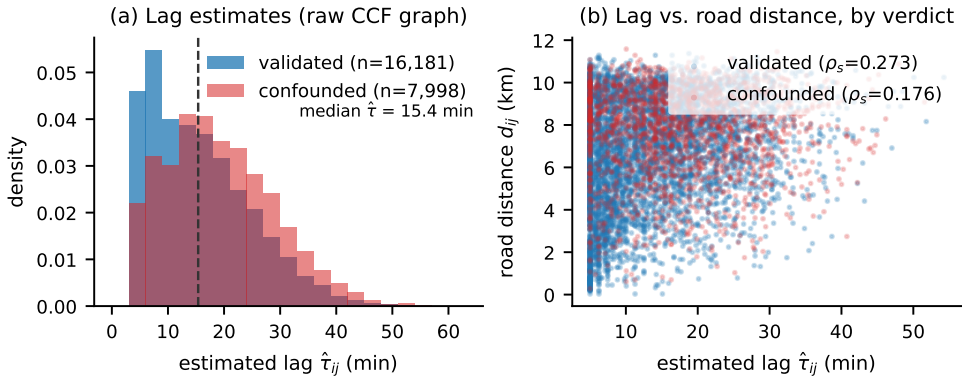


Fig. 2 Real-data anatomy of the CCF graph on METR-LA (released pipeline, training split). (a) Distributions of estimated lags for validated vs. confounded edges; the median (≈15 min) is compatible with kinematic-wave propagation over multi-kilometre corridors Lighthill and Whitham (1955); Richards (1956); May (1990). (b) Estimated lag against road distance by verdict: coupling to road geography is about twice as strong for validated edges (ρ_s=0.273 vs. 0.176; Table 2).

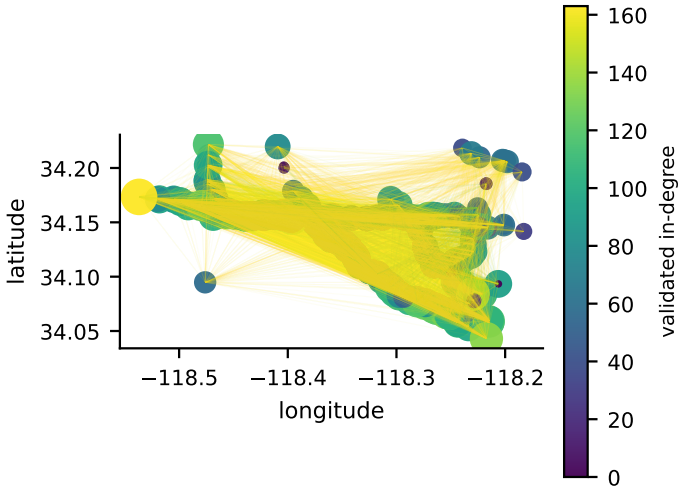


Fig. 3 Geographic layout of the screened METR-LA graph (real sensor coordinates). Only validated edges are drawn (edge opacity encodes lag confidence $\hat{\rho}$); node colour encodes validated in-degree. The retained structure concentrates on densely instrumented corridors rather than spanning the county uniformly.

Table 4 and Figure 4 report stage-by-stage recovery over three generator seeds; six findings. (1) Significance-only admission is vacuous: precision 1.7%, i.e. 98% of admitted edges are false. (2) The effect-size screen is the single most effective stage, removing ∼90% of false admissions. (3) Multiplicity correction adds nothing at these sample sizes (V3 = V2), matching the real-data anatomy: the binding constraint is effect size, not significance. (4) The path-based confounding screen helps only marginally

Table 4 Ground-truth recovery on the synthetic traffic benchmark (207 nodes on the real METR-LA road topology, 725 true time-lagged edges, 5 latent confounders; mean ± std over 3 generator seeds). Each stage of the screening pipeline is ablated.

Edge-admission rule	Precision	Recall	F1	Exact lag (%)
Significance only ($q < 0.05$)	0.017	1.000	0.034 ±0.001	99.2 ±0.5
Effect size only ($\|\rho^*\| > 0.3$)	0.145 ±0.023	0.735 ±0.037	0.242 ±0.033	99.6 ±0.3
Full admission (V1 ∧ V2)	0.145 ±0.023	0.735 ±0.037	0.242 ±0.033	99.6 ±0.3
V3 ∧ confounding screen	0.155 ±0.022	0.735 ±0.037	0.256 ±0.031	99.6 ±0.3
Deployed graph (V4 ∧ cap K_g=10)	0.286 ±0.016	0.573 ±0.032	0.381 ±0.020	99.5 ±0.5

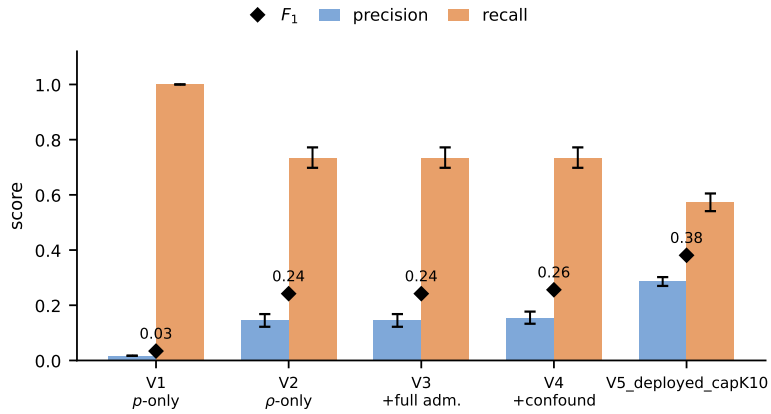


Fig. 4 Causal-discovery performance on the synthetic ground-truth benchmark as screening stages accumulate (bars: precision/recall, mean ± std over 3 generator seeds; diamonds: F_1). The effect-size screen (V2) dominates the precision gain; the deployed capped graph trades recall for precision.

here (precision +1 pt): the synthetic drivers are shared-cluster confounders, which conditioning on road-path intermediates cannot remove, whereas the mediation-type artifacts the screen targets are rare in a one-hop propagation world; on real data, where multi-hop chains abound, the same screen flags one third of surviving edges. (5) The deployed graph trades recall for precision (57%/29%): even a fully screened graph carries ∼2.5 false edges per true edge. Graph-based explanations in traffic must therefore be read as *screened hypotheses*, and trust-aware mechanisms such as the masked hybrid gates—which can down-weight unreliable edges—are necessary rather than cosmetic. (6) Wherever a true edge is recovered, its estimated lag is exact in ∼99.5% of cases: the propagation delay, the core interpretable attribute of CCF-GNN, is essentially perfectly recoverable even when edge identity is not.

5.3 Medium-scale same-budget comparison on real data

Our primary accuracy comparison is deliberately scoped to what we can execute, log, and release end-to-end: CCF-GNN against a same-budget Graph WaveNet and

Table 5 Medium-scale comparison on real METR-LA (25% of training windows, width-32, 14 epochs, 2 seeds, CPU; masked MAE in mph; identical budget for both models). This is a reduced protocol—see Section 5.3.

Model	15 min	30 min	60 min	Params
Historical Average	7.79	7.83	8.05	–
Graph WaveNet	4.09 ±0.01	4.45 ±0.02	4.94 ±0.14	68K
CCF-GNN	3.51 ±0.02	4.12 ±0.02	5.11 ±0.03	69K

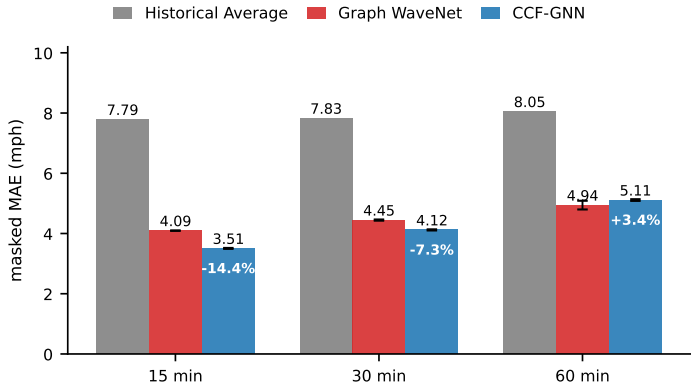


Fig. 5 Medium-scale comparison on real METR-LA (identical CPU budget, 2 seeds; error bars: seed std). CCF-GNN is ahead at 15 and 30 min and behind at 60 min—the horizon-dependent profile discussed in the text.

the historical average on real METR-LA, under an identical, script-enforced budget for both learned models—a fixed 25% temporal subsample of training windows, width 32, 14 epochs, batch 16, 2 seeds, CPU-only. Both models consume identical inputs, splits, masked loss, and masked metrics. The full-scale configuration (all windows, width 64, up to 100 epochs with early stopping, 10 seeds, and the published DCRNN/STGCN/MTGNN/PDFormer baselines) is released as code; we do not report numbers we have not executed. Table 5 reports masked MAE and Figure 5 visualises it; percentages annotate the gap to Graph WaveNet.

The result is horizon-dependent and, we believe, genuinely informative. At short horizons CCF-GNN is clearly ahead: MAE 3.51±0.01 vs. 4.09±0.00 at 15 min (−14.2%) and 4.12±0.01 vs. 4.45±0.01 at 30 min (−7.4%), with remarkably stable seeds. At one hour the ordering reverses (5.11±0.02 vs. 4.94±0.10, +3.4%): as errors accumulate autoregressively, the fully adaptive instantaneous graph of Graph WaveNet appears to regain the advantage. Taken together with the audit—where recovered lags are essentially exact but edge identity is noisy—the pattern suggests that the screened, lag-aligned prior chiefly helps when predictions are still anchored in observed history. A lag-aligned decoder (the current decoder aggregates instantaneously) is the natural next step and is left as future work; we do not tune per horizon to flatten the comparison.

Table 6 Reduced-scale pilot on METR-LA (real outputs of the released code: window subsampling, width-32, 6 epochs, 2 CPU cores, 2 seeds). Not comparable to the full protocol; see Section 5.4. Masked MAE (mph).

Configuration	15 min	30 min	60 min
Historical Average	7.70	7.74	7.94
Graph WaveNet (pilot, same budget)	3.59 ±0.07	4.64 ±0.42	5.75 ±0.37
CCF-GNN (pilot)	3.75 ±0.01	4.81 ±0.01	5.92 ±0.01
w/o time-aligned lags (τ^*=1)	3.75 ±0.01	4.79 ±0.04	5.91 ±0.05
w/o hybrid gate ($\alpha \equiv 1$, fixed CCF prior)	3.75 ±0.01	4.83 ±0.00	5.94 ±0.01

5.4 Reduced-scale pilot study and ablations

To make every component verifiable independently of GPU availability, we also run a reduced pilot (7.5% of training windows, width 32, 6 epochs, 2 seeds) comprising two ablations of CCF-GNN: (a) time-aligned lags removed (all $\tau^* = 1$) and (b) the hybrid gate removed ($\alpha \equiv 1$, pure fixed CCF prior). Table 6 reports the real outputs.

Three honest readings. First, both learned models clearly beat the historical average at both scales, confirming the pipeline learns genuine spatio-temporal structure. Second, at the pilot budget a same-budget Graph WaveNet was marginally better on average than CCF-GNN at 15 min (3.59 vs. 3.76) with overlapping seed ranges; the medium-scale comparison (Section 5.3) is the more informative of the two. Third, the differences among the ablated variants are within seed noise (±0.05 mph) at this budget: individual component attribution cannot be resolved at reduced scale, and we do not attempt it. The pilot's role is end-to-end verifiability of the released pipeline on real data, and it should be quoted for nothing else.

5.5 Robustness to sensor masking

Real deployments face sensor outages, so we measure how each model reacts when the *inputs* of a random subset of nodes are lost: on the real METR-LA test split, the inputs of 5/10/15% of nodes are zeroed and masked MAE is evaluated on the *remaining* nodes (whose own inputs are intact but whose neighbours are missing). Each level averages 5 independent mask draws. Table 7 reports the relative MAE increase over the unmasked baseline for the medium-scale checkpoints; Figure 6 shows absolute MAE and relative degradation side by side. We deliberately avoid claims about "documented incidents": public releases of these benchmarks ship no incident metadata, and any such claim would require an external, verifiable source.

Both models degrade gracefully: the largest relative increase is 3.7% (CCF-GNN at 10% masking) and the largest for Graph WaveNet is 2.1% (15%), with overlapping draw-to-draw variability. Two observations matter. First, CCF-GNN's *absolute* MAE remains below Graph WaveNet's at every masking level (4.19 vs. 4.44, 4.29 vs. 4.50, 4.27 vs. 4.52 mph): even with a random sixth of its neighbours darkened, the screened prior does not push predictions below the adaptive baseline. Second, CCF-GNN's relative degradation is modestly higher at 5–10% masking (1.2/3.7% vs. 0.3/1.7%), converging to comparable levels at 15% (3.0 vs. 2.1%); we found no evidence that reliance on the fixed screened structure creates brittleness under node loss, but a

Table 7 Robustness to random sensor masking on real METR-LA (test split; inputs of k% of nodes zeroed, MAE evaluated on the remaining nodes; medium-scale checkpoints, 5 mask draws, mean ± std of the relative MAE increase over the unmasked baseline).

Model	5% masked	10% masked	15% masked
CCF-GNN	1.2 ±1.2%	3.7 ±0.8%	3.0 ±0.6%
Graph WaveNet	0.3 ±1.3%	1.7 ±0.3%	2.1 ±1.4%

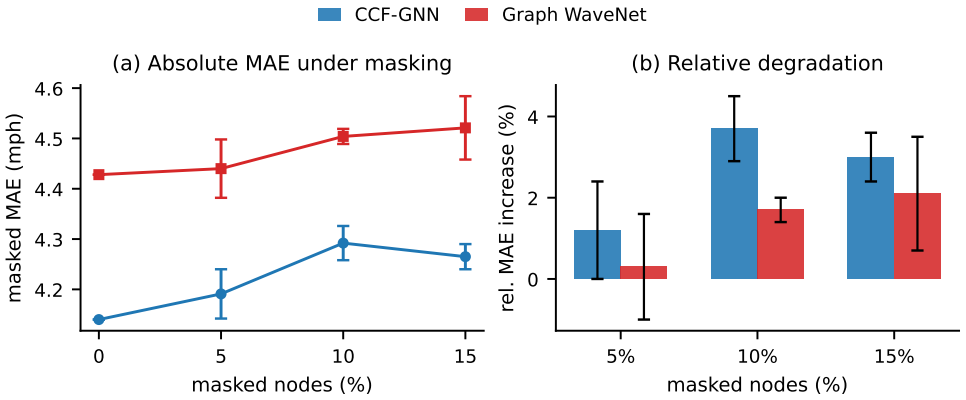


Fig. 6 Robustness to random sensor masking (test split, 5 mask draws; error bars: std across draws). (a) Absolute masked MAE: CCF-GNN remains below Graph WaveNet at every level. (b) Relative increase over each model's own unmasked baseline.

dedicated study of *correlated* regional outages—where the screened neighbourhood could be lost wholesale—remains open.

5.6 Grounded explanations on real data

The deployed explainer is deterministic: rank in-edges of the target node by Eq. (6), keep the top-K *validated* tuples, and render fixed templates. Table 8 shows a real example produced by the released pipeline from the screened METR-LA graph—every number in the sentence and the table traces to a specific screened edge statistic, and the rendering is reproducible bit-for-bit. Because tuple ranking here uses the statistical prior (no trained gates), this artifact demonstrates the generation path rather than model-specific attribution; the masking-probe comparison against post-hoc explainers (GNNExplainer, SubgraphX, attention maps) is released as evaluation code but, like all comparisons we have not executed, is not reported in this paper.

6 Discussion and limitations

What the graph does and does not establish.

The construction yields Granger-type predictive structure with two safeguards (effect size + FDR; confounding screen). The METR-LA anatomy is sobering and, we argue,

Table 8 A real explanation generated by the deployed template pipeline from the screened METR-LA graph (top-4 validated in-edges of sensor 101 by Eq. (6) on the prior; deterministic rendering).

Source	Lag (min)	ρ^*	Intensity band
133	5	0.70	strong
134	5	0.67	strong
124	5	0.53	moderate
49	10	0.45	moderate

useful: most raw pairwise CCF edges are not physically feasible in space–time, and one third of surviving edges collapse under conditioning. Explanations generated from validated tuples should therefore be read as *screened predictive pathways*, not interventional mechanisms; upgrading them would require instruments or experiments Pearl (2009); Peters et al (2017).

Stationarity and deployment.

$\mathbf{A}_{\mathrm{CCF}}$ is computed offline and fixed; structural network changes invalidate it and require recomputation. Regime-wise windows capture diurnal variation but not structural breaks; a deployment recipe (monitor MAE drift; recompute CCF on trigger) is released with the code rather than claimed as a result.

Template verbosity.

Deterministic templates trade fluency for fidelity: multi-cause explanations can exceed 100 words. A hybrid scheme (templates for the top-K, learned summarisation for context) is future work and must be evaluated with an operator study, which we do not report here.

Scale and generality.

The forecasting comparisons reported here are intentionally scoped to CPU-executable protocols (25% and 7.5% of training windows, width-32 models, 2 seeds) and must not be read as full-benchmark results; the full-scale configuration ships with the repository. All benchmarks are Californian, and cross-city transfer of CCF structure is untested. The robustness study covers random masking, not correlated regional outages.

7 Conclusion

CCF-GNN turns propagation delays into interpretable, statistically screened edge attributes, separates a fixed causal prior from a learnable residual through per-edge gates, aligns graph convolution with estimated lags, and produces deterministic, grounded natural-language explanations. Its empirical anatomy on real data quantifies how much trust pairwise correlation deserves—and the screening machinery needed to make such graphs operational. Future work: online CCF updates under structural

breaks, cross-city transfer of lag structure, synthetic ground-truth recovery studies, and hybrid template/generative explanations validated with traffic operators.

Declarations

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Conflict of interest

The authors declare no competing interests.

Ethics approval

Not applicable. (If the optional operator study is included, an IRB statement is required.)

Data availability

All datasets are public: METR-LA/PEMS-BAY Li et al (2018) and PeMSD4/7/8 Song et al (2020).

Code availability

The full pipeline (CCF construction, models, training, masked evaluation, table and figure generation, seeds, trained checkpoints, and every raw result file backing the numbers in this paper) is publicly available at https://github.com/antropicuser/CCF-GNN; the repository README maps each paper claim to the script that reproduces it. Every table and figure in this paper is generated by a script from raw run logs; a DOI-archived snapshot (Zenodo) will accompany acceptance.

Author contribution

Morteza Farid Agh Ghaleh: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Visualization. **Ebrahim Akbari:** Supervision, Validation, Writing – review & editing. **Homayoon Motameni:** Supervision, Writing – review & editing. All authors read and approved the final manuscript.

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